

GEILENKIRCHEN, Germany

WMO: 105000

Lat: 50.9592N

Lon: 6.0392E

Elev: 251

StdP: 14.56

Time Zone: 1.00 (EUC)

Period: 90-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 17.6	(c) 21.4	(d) 12.6	(e) 10.6	(f) 20.1	(g) 16.5	(h) 12.8	(i) 23.5	(j) 27.5	(k) 47.2	(l) 24.9	(m) 45.7	(n) 5.2	(o) 90	(p) 0.516

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	87.5	67.4	83.7	66.2	79.8	64.9	69.3	82.8	67.6	79.7	66.1	77.6	7.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	92.5	73.7	63.0	87.0	72.1	61.6	82.6	70.8	33.5	82.8	32.1	79.5	30.9	77.6	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 21.7	(b) 18.8	(c) 16.5	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 10.5	(b) 73.3	(c) 7.4	(e) 2.2	(f) 5.2	(g) 74.9	(h) 0.9	(i) 76.2	(j) -3.2	(k) 77.4	(l) -8.6	(m) 79.1	(n) 79.1	(o) 79.1	(p) 79.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.2	38.0	40.1	45.0	49.5	56.6	61.3	66.3	65.7	58.8	51.3	42.8	38.6
	DBStd	12.11	8.57	8.65	6.08	6.79	6.94	5.88	5.76	5.93	5.04	6.66	7.49	7.61
	HDD50	1599	376	284	178	91	17	0	0	0	1	66	230	356
	HDD65	5256	837	696	620	465	274	144	53	65	195	425	665	817
	CDD50	2055	4	8	24	77	222	338	504	486	266	107	15	4
	CDD65	237	0	0	0	1	14	33	92	86	10	1	0	0
	CDH74	2436	0	0	1	17	172	367	937	856	79	7	0	0
	CDH80	778	0	0	0	0	32	103	318	308	17	0	0	0
Wind	WSAvg	6.8	8.3	8.3	7.8	6.6	5.8	5.9	6.0	5.6	6.1	6.6	6.8	7.9
Precipitation	PrecAvg	32.0	2.7	2.6	2.2	1.9	2.4	3.0	3.1	3.3	2.5	2.5	2.7	3.1
	PrecMax	38.9	5.4	6.0	4.5	3.8	5.1	6.3	7.3	6.7	6.2	6.0	4.5	6.9
	PrecMin	25.7	0.2	0.4	0.2	0.0	0.9	0.7	1.2	1.0	0.6	1.3	0.2	0.9
	PrecStd	3.7	1.3	1.3	1.1	0.9	1.0	1.3	1.6	1.7	1.5	1.0	1.1	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.2	62.2	66.5	77.0	84.4	88.5	92.4	92.2	82.9	74.3	60.0	56.2
		MCWB	50.5	50.1	53.6	56.6	65.0	69.7	69.8	67.6	63.7	63.0	53.7	51.9
	2%	DB	53.9	57.2	61.9	70.2	78.6	82.5	87.5	87.7	75.4	68.0	56.8	53.9
		MCWB	49.3	49.2	51.5	56.5	63.4	66.0	68.0	66.8	62.4	59.8	52.6	50.1
	5%	DB	51.7	54.0	58.2	65.9	74.6	78.2	83.3	83.4	71.0	64.5	55.1	51.8
		MCWB	47.2	48.1	50.2	54.2	60.7	64.4	66.3	66.2	61.0	57.4	51.5	48.7
	10%	DB	49.3	51.2	55.0	61.2	69.9	73.3	79.3	78.7	67.7	61.3	52.8	49.4
		MCWB	45.8	46.6	49.0	52.0	58.9	61.8	65.3	64.9	59.5	56.2	49.5	46.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.3	52.2	54.9	61.0	67.1	70.3	72.2	71.1	66.4	63.2	55.2	53.1
		MCDB	55.4	58.6	63.5	70.7	79.6	84.3	87.0	86.0	75.9	70.3	57.9	55.4
	2%	WB	49.8	50.4	52.7	57.5	64.1	68.0	69.8	68.5	64.0	60.8	53.6	50.8
		MCDB	53.0	55.7	58.9	66.7	75.6	79.9	83.2	81.5	72.3	66.8	56.4	53.4
	5%	WB	47.9	48.5	51.1	55.0	61.9	65.4	67.9	67.2	62.1	58.7	51.8	48.8
		MCDB	51.3	53.1	56.5	63.4	71.9	75.2	80.3	79.5	68.5	63.2	54.5	51.7
	10%	WB	45.7	46.9	49.7	52.9	59.4	63.1	66.0	65.7	60.5	56.4	49.9	46.3
		MCDB	49.0	50.9	54.8	61.0	68.0	71.6	77.1	77.2	66.6	60.8	52.2	48.9
Mean Daily Temperature Range	5% DB	MDBR	9.6	11.0	14.0	16.6	17.7	17.2	18.7	19.0	14.8	13.4	9.9	8.5
		MCDBR	11.9	14.5	20.2	23.4	25.4	25.9	27.3	27.5	20.3	17.6	11.0	10.3
		MCWBR	9.3	9.6	12.3	11.9	12.7	12.8	12.0	11.4	10.5	11.1	8.5	8.9
	5% WB	MCDBR	10.9	12.5	16.1	20.2	23.4	22.6	23.3	23.4	17.7	16.5	10.6	9.7
		MCWBR	9.2	9.4	10.6	11.1	12.4	12.2	11.0	11.2	9.7	11.4	8.8	9.0
Clear-Sky Solar Irradiance	taub		0.323	0.346	0.378	0.413	0.419	0.420	0.425	0.416	0.387	0.362	0.337	0.318
	taud		2.411	2.367	2.285	2.211	2.233	2.250	2.236	2.285	2.353	2.416	2.409	2.382
	Ebn at Noon		222	246	258	262	266	266	262	258	253	237	213	202
	Edh at Noon		20	26	34	41	42	42	42	38	32	25	20	17
All-Sky Solar Radiation	RadAvg		239	455	816	1305	1556	1693	1603	1369	1016	597	280	185
	RadStd		35	88	121	149	144	151	193	133	114	54	34	26

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (84 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page